Rakib - AI Helmet for Safety

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Final Report

FINAL REPORT

For

Rakib - AI Helmet for Safety

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Abstract

Motorcycle delivery riders in Qatar face heightened safety risks due to fatigue, long working hours, and environmental extremes. Recent trends in Qatar indicate increased safety challenges for the delivery riders, and traditional helmets offer only passive protection, while these smart helmets focus mainly on communication or navigation, but not on physiologically monitoring the responses. To address this gap, we present Rakib, an AI-enhanced smart helmet designed to prevent, detect, and respond to rider incidents in real time.

Rakib uses a MAX30102 PPG/HRV sensor for tracking heart rate variability and SpO_2 for fatigue detection, along with an MPU-6050 accelerometer/gyroscope to identify impacts or abnormal motion. A LilyGO T-SIM7600G-S3 microcontroller used for data processing which runs TensorFlow Lite models which ensures low-latency alerts. Upon detecting fatigue, the system triggers visual and audible warnings. In the event of a collision, an automated SOS alert with GPS coordinates is transmitted to emergency services via an integrated LTE/GNSS module.

A mobile application, integrated with a Firebase cloud backend, enables comprehensive ride analytics, performance tracking, and fleet management insights. The system incorporates six key subsystems—sensor, microcontroller, communication, power, alert interface, and information storage—all seamlessly embedded within the AGV K1-S helmet shell without compromising safety, comfort, or balance. Key technical challenges, including power management, sensor accuracy, and connectivity reliability are addressed through a rechargeable Li-Po battery. That includes an integrated battery management system, cross-sensor validation to minimize false readings, and robust LTE/GPS integration for consistent SOS transmission.

Benchmarking against commercial smart helmets confirms Rakib occupies an exclusive position by providing integrated AI-driven fatigue detection with standalone emergency alerting. The project applies principles from circuit theory, digital system design, and electronics learned in the classes ECEN 214, ECEN 248, and ECEN 325, respectively. Besides that, Rakib follows state-of-the-art standards for impact safety ASTM F1446 and data security ISO/IEC 27001. Preliminary validations of hardware components have already confirmed the functionality of sensors and core alert mechanisms. Further tasks to be performed include refining the AI model, optimizing energy, collection of real data, and safety certification. Rakib presents a technologically advanced solution that is socially responsive to improving rider safety in Qatar's Delivery Sector.

Chapter 1: Concepts of Operation (ConOps)

1.1 Scope

The project thus involves a system that is being developed to authenticate the core functions and operational performances of Rakib. The helmet uses artificial intelligence and sensors to further improve motorcycle riders' safety through real-time fatigue and crash monitoring. This is a significant step forward in potentially reducing accident risks and improved riding conditions through efficiently providing SOS alerts and data insights into safer transportation. The Rakib Helmet system continually collects rider data, analyzes it, and monitors for fatigue and crashes. Rakib monitors rider fatigue by measuring heart rate and SpO_2 using a MAX30102 PPG/HRV sensor, while crash detection is achieved through an MPU-6050 accelerometer and gyroscope that track impact forces and motion. The system is built around a LilyGO T-SIM7600G-S3 microcontroller (ESP32-based), which processes data from both sensors in real time to enable timely detection and alerts. In the event fatigue is detected and a crash occurs, a buzzing sound will be created and it informs the rider that SOS will be sent, it will generate an SOS message with the GPS location through the LTE/GNSS module to the emergency services. The mobile app saves the entire ride data. There are currently no helmets that synergistically prevent, detect, and respond as one integrated system. This gap demonstrates the urgent need for a smart helmet that monitors rider safety for fatigue and provides immediate alerts when needed. The Rakib helmet addresses that demand by integrating fatigue monitoring, crash detection, and automatic emergency response into a single device.

1.2 Operational Description

The Rakib Helmet consists of a real-time fatigue monitoring system that detects crashes and communicates with emergency services; it constantly records and processes motion and physiological data from various sensors with the help of a microcontroller. The major reason lying behind the design of Rakib is that it is meant to be used on a daily basis under suburban and urban environments, which is the reason for the diverse conditions that the helmet can withstand, such as temperatures ranging between -10 degrees Celsius and around 55 degrees Celsius, moderate dust and humidity, levels of vibrations and motions during motorcycle rides, and areas with sufficient connection of LTE/GPS signal strengths to send the SOS messages and alerts. These environmental conditions are required to ensure the stable operation of the AI Voice assistant, data logging, and ride analysis which are provided by mobile data or Wi-Fi connections.

However, the electronic system is not designed to operate under extreme environmental conditions such as floods or severe storms. The helmet shell itself maintains its structural integrity and safety certification within its intended use. AI fatigue and crash detection accuracy

can be affected by sensor location and fit of the helmet. The helmet is intended for single-user operation to maintain sensor calibration accuracy, as its AI and sensors are adapted to each rider's personal physiological and behavioral data, so that accuracy will be affected if another person were to use it.

1.3 System Description

The Rakib AI Helmet incorporates several subsystems integrated with each other to form an intelligent safety network that works before, during, and after a motorcycle accident, and each subsystem collects, processes, and forwards important data ensuring rider safety through fatigue warning systems and real-time crash detection and emergency response systems. All the electronic components applied for this are incorporated within the AGV K1-S helmet itself in a manner that ensures the added hardware does not affect helmet comfort, balance, and ventilation while maintaining its protection and safety certification.

1.3.1 Sensor Subsystem

The sensor part of the subsystem collects both physiological and motion data to monitor the fatigue of the riders and identify crashes in real time, while the sensors are interfaced to the microcontroller that processes the signals to identify any abnormal conditions. It uses MPU-6050 (Accelerometer/gyroscope) Sensor to measure the linear acceleration and angular velocity, measure along the X, Y, and Z axes, and it will detect sudden shocks, tilt, and unusual head movements as associated with a crash. Installed directly on the topmost center of the helmet for balance and precise measures, the MAX30102 (PPG/HRV) Sensor determines heart rate, heart rate variability, SpO_2, and pulse to probe fatigue, stress, and any unusual physiological responses. The LilyGO T-SIM7600G-S3 (ESP32) will receive and analyze the information from each of the sensors to detect crashes or fatigue, and manage communication and alerts.. These sensors collectively support system operation by using some functionalities like collection of data from the sensors and transmission to the app, integration of sensing, computing, and communication functions, Power management and data logging, and app / cloud communication with LTE, GPS, and Bluetooth.

Figure 1: Accelerometer (left), PPG sensor(middle) and ESP32 Microcontroller(right)

1.3.2 Communication Subsystem

The Communication Subsystem manages all connectivity between the helmet and external devices. It enables automated SOS alerts via LTE with GPS location data, while Bluetooth

maintains a connection to the app for user interaction and voice assistance. The design requires positioning earphones closer to the rider's ears and near their mouth to ensure that speech is captured effectively. The LTE Automatically send SOS messages to live GPS coordinates once or when a crash is detected, GPS has provided a precise location for emergency response quickly, the Bluetooth connects the helmet with the Rakib app for transferring the data and the AI assistant and Speaker and Microphone (Sena Kit) enable two-way communication with the AI voice assistant through the Rakib app as it is hands-free voice commands and responses during rides.

Figure 2: Sena Microphone and Speaker

1.3.3 Power Subsystem

The power subsystem guarantees the trusted operation of all electronic components and simultaneously ensures the balance, safety certification and comfort of the helmet. To achieve all these goals, it uses a voltage regulator, rechargeable Li-Po battery and a BMS (battery management system). It is positioned at the back of the helmet under inner padding for protection and stability. Other features include an ultra-light Li-Po battery with USB-C charging port and protection circuits against overcharging and overcurrent, including short circuits. It is powered by a microcontroller, which constantly monitors battery status.

Figure 3: Rechargeable Li-Po battery

1.3.4 Alert and User Interface Subsystem

The entire subsystem for alerts and user interface, completely ensures that visual, sound, and voice interaction is continually made available to a rider and the emergency contacts. It ensures continued alerts in the event that the network suffers downtime. The RGB LED Stick uses a color code to report system status (green = SOS sent, red = SOS not sent, blue = fatigue detection). Situated above the visor of the helmet to enable early detection and avoid distractions,

it alerts the rider, using a countdown test, through a countdown buzzer sound, before sending the SOS message, thus giving time for the rider to hear the warning and cancel, if necessary. Given as a sound alert, Push Button comes as an option for riders to cancel SOS or fatigue alerts manually. This component is positioned within the helmet. This is mounted on the side of the helmet for the rider to manually cancel the SOS function if needed and app interface to log fatigue level, crash alerts, battery status and also receives and transmits ai feedback.

Figure 4: RGB Led stick (left), Pierzo buzzer(middle) and Pushbutton (right)

1.3.5 Information Storage and Analytics Subsystem

The subsystem logs sensor data, fatigue events, and crash history into the Rakib Cloud database. The stored data enhances AI performance and enables users to view ride analytics via the mobile app dashboard. It gets ride data and SOS alerts from the helmet via the LTE Communication subsystem, which stores them for future analysis. The app works with the AI to do the analysis and updates the status by communicating the microcontroller via bluetooth, displaying ride history, fatigue alerts, and safety logs for features that help gauge rider performance and improve fatigue detection accuracy over time.

1.4 Modes of Operation

Rakib will have a basic functioning mode with data collected from sensors, processed by a microcontroller, and stored for analysis. The helmet will keep track of rider vitals and motions continuously with all the readings being normal. The system will contain an alert mode also activated on the detection of fatigue. In case of any accident, it switches to emergency mode. After every ride, the helmet will be in data upload mode. Finally, transferred information is kept and analyzed using the Rakib mobile app and the cloud. The uploaded data will be available for review and analysis on performance evaluation and future improvements.

1.5 Literature and Learning Resources

Our group conducted extensive research to identify the technical requirements and know the feasibility of the application. The references in our Concept of Operations (ConOps) report were mainly scholarly articles and journals which are credible and are a technical foundation of the project. The research includes real-world statistics on delivery rider accidents and fatigue-related

injuries. Benchmark for existing smart-helmet crash detection, highlights lack of AI and fatigue monitoring; comparison of communication and safety features in commercial helmets.

Chapter 2: Benchmarking

2.1 Problem Statement and Proposed Solution

2.1.1 Problem Statement

Riders in Qatar experience significant fatigue, particularly during high-temperature conditions and long working shifts. Long hours without any rest could be potential hazards with a corresponding delay in any emergency response. Traditional helmets provide only passive protection during impact, but do not avert accidents. Smart helmets available on the market are more oriented toward communications or navigation functions. Although these add value to the market, they do not offer any proactive fatigue monitoring or automatic emergency response for the rider. The objective of this project is to develop a smart helmet that ensures rider safety by detecting fatigue and accidents early to intervene in accidents and supply relevant emergency post-ride information.

2.1.2 Proposed Solution

The design of the Rakib AI Helmet keeps use of artificial intelligence and smart sensors so that it can identify and alert when the rider is not alert and in the event of a crash. The sensor MAX30102 (PPG/HRV) and MPU-6050 (Accelerometer/gyroscope) will be used to measure the heart rate variability, SpO_2 and impact on the rider. The helmet has a microcontroller system LilyGO T-SIM7600G-S3 (ESP32) for the processing of real-time data collected and, in case the system is offline, local fatigue alerts will be made through the LED indicators while online the SOS and fatigue alert system will get activated through the system to communication when the helmet is online. In case of a crash, the LTE/GNSS module will instantly transmit to emergency services the SOS alert together with the rider's location live and later stored in the Rakib mobile app. Post ride data is uploaded in the Firebase cloud and mobile app and enables users and fleet managers to continuously analyze fatigue patterns and riding behavior. The Rakib Helmet is not just safe for the riders and reduces risk of accidents. It is actually a proactive step taken to make delivery operations safer within Qatar.

2.2 Existing Solutions

2.2.1 Quin Smart Helmet

Quin Design smart helmet is based on emergency alert features reliant on crash detection and Bluetooth connectivity from its application connected on a smartphone. It boasts benefits such as

hands-free communication, attractive design, and sensors embedded for impact detection. However, it has several limitations in that it cannot sense physiological fatigue and is mostly reliant on smartphone connectivity, not being AI-driven and solely limited to smartphone-based SOS.

2.2.2 Sena Smart Helmet

In the order of giving importance to multimedia and intercom functions, Sena prioritizes multimedia and communication features, A certain model has voice control and even an optional crash sensor. It has a reliable intercom as well as very good communication; its performance in group riding convenience is limited compared to those of more commercial brands, but it has very little in the way of safety analysis, consumes extra power, doesn't cut age, and doesn't combine with a cloud. More targeted for the non-professional rider.

2.2.3 Forcite MK1 Helmet

Forcite does not keep track of physiological measures or fatigue, but it has cameras and navigation lights, and it has effective communication. It has a built in HD camera, and it is strong and high quality materials but SOS must be manually activated, and there are no AI or fatigue models available and it targets only the higher end market and has very high production costs.

2.2.4 LIVALL Smart Helmet

The LIVALL helmets are designed to connect with any smartphone for Bluetooth distress alerts and fall detection. They can be considered as smart helmets relatively low-cost. Also, compared with others, they are light in weight and have voice communication. However, the helmets lacks physiological monitoring capabilities on fatigue data, have limited AI processing, and show poor impact detection accuracy, not to mention the fact that they must always be connected via the smartphone app.

2.2.5 Proposed Rakib AI Helmet

This essentially incorporates crash-detection, fatigue monitoring and LTE/GNSS based SOS with cloud analytics. Initially, a prototype is much more costly than commercially available helmets. Also, AI model training and data set collection are needed.

2.3 Benchmarking Criteria Comparison

Benchmarking compares Rakib's performance with existing products according to engineering and social conditions such as public health and welfare, as well as environmental and economic considerations.

2.3.1 Public Health

- Enhances rider safety by preventing fatigue-related accidents.

- Provides AI-based crash detection for proactive protection.

2.3.2 Public Safety

- Ensures timely emergency alerts through LTE/GNSS SOS transmission.
- Maintains secure data handling for rider information.

2.3.3 Public Welfare

- Builds confidence, comfort, and reliability among riders.
- Supports peace of mind for delivery workers and fleet managers.

2.3.4 Cultural & Social Factors

- Adapts to Qatar's delivery culture and long working shifts.
- Supports local working conditions, addressing fatigue challenges.

2.3.5 Global & Environmental Factors

- Promotes **road safety sustainability**.
- Reduces **injury-related social costs** by preventing accidents.

2.3.6 Economic Factors

- Evaluates affordability of the prototype (target below USD 1000).
- Considers scalability for widespread adoption.
- Assesses market entry potential in Qatar and beyond.

2.4 Benchmarking Table

Table 1: Benchmarking Table

Feature / Metric	**Quin Helmet**	**Sena Helmet**	**Forcite MK1**	**LIVALL Helmet**	**Rakib AI Helmet (Proposed)**
Fatigue Detection (HRV/IMU)	NA	NA	NA	NA	A
Crash Detection (AI-Based)	A (via Smartphone)	A	A	A (basic)	A (on-device)

Automatic SOS (w/ GPS)	A (phone required)	NA	NA	A	A (LTE/GNSS)
On-Device AI Processing	NA	NA	NA	NA	A (ESP32 + TensorFlow Lite)
Data Analytics / Cloud Storage	NA	NA	NA	NA	A (Firebase + App)
Voice / User Interface	A	A	A	A	A
Public Health Impact	Moderate	Moderate	Low	Moderate	High
Power Consumption	Medium	High	High	Low	Medium (optimized)
Cost (Approx. USD)	700	600	1000	200	900
Availability	USA	Global	Australia	Global	Local Prototype (Qatar)
Overall Safety Level	Medium	Medium	Medium	Low	Very High

NA – Not Available, **A** – Available

2.5 Benchmarking Summary

There is the distinct quality of Rakib, where the comparables are involved like Forcite, Sena, LIVALL, Quin. Even an average usage of most of these helmets is based on a kind of post-crash model. Rakib, the main project idea, initiated a pre-crash paradigm for fatigue detection and alert before a crash. Many practical ideas will accompany this study in improving the design, like using LIVALL-style light materials for more comfortable wear, applying Sena's battery management methods to improve power life, and bringing in Quin's mobile app design-inspirational improvements to make the Rakib app more user friendly-all inspired by Forcite's padding and ventilation design, all combined produced a benchmarking process that was clearer to the design of Rakib, helped in differentiating Rakib to be one of the first helmets to integrate AI-based fatigue detection and crash alerts with SOS communication. Benchmarking proved the Rakib Helmet project to be heading in the right direction. It is clear from the other helmet features that Rakib is different as it transcends most of them by including AI and built-in SOS features to highlight their cases of prevention and response. The results suggest that the concept is indeed feasible, necessary, and highly beneficial, particularly for delivery riders in

Qatar. The design of the system was also upgraded regarding user comfort as well as power management. Future work will focus on enhancing energy efficiency, collecting additional real-world data to improve the models and moving towards certification for market entry as a safe and reliable helmet for consumers.

Chapter 3: Functional System Requirements (FSR)

3.1 Systematic Overview & Scope

The Rakib Helmet aims at enhancing motorcycle delivery rider safety through real-time crash and fatigue event detecting features. It provides early detection and rapid-response emergency safety measures against high accident risks among delivery riders. The scope of the project is to develop a completely functional prototype of the Rakib Helmet that can be used for several hours. This prototype will accurately detect fatigue and crash events and initiate communication alerts through artificial intelligence. The model will be powered using a lithium battery which enables several hours of continuous system operation; therefore, conserving energy. The main brain of the system is the LilyGO T-SIM7600G-S3 (ESP32) microcontroller which has a built-in bluetooth and LTE/GNSS module. It has heart rate monitoring and oxygen levels as well as monitors impact force effects using the MAX30102 (PPG/HRV) sensor and the MPU-6050 (Accelerometer/gyroscope) sensor. The data collected from sensors are processed by the ESP32 and will be passed on to the AI model for analysis to check for fatigue signs or crash events.

3.2 Systematic Requirements

In this section, "Rakib AI Helmet for Safety" refers specifically to the complete system being developed as proof of concept, including the built-in sensors in the helmet, main microcontroller, power and alert components, communication modules, and the online database where all data will be stored and displayed.

3.2.1 System Definition

Six subsystems form the Rakib helmet: Sensor, Microcontroller, Communication, Power, Alert and User Interface, Information Storage and Analysis. Each part serves its purpose. Together, they complete a safety system that detects collisions, rider fatigue, and communicates some important data to external systems for real-time response, while collecting and sharing data in an efficient manner, without compromising the rider's stability, comfort, and safety standards.

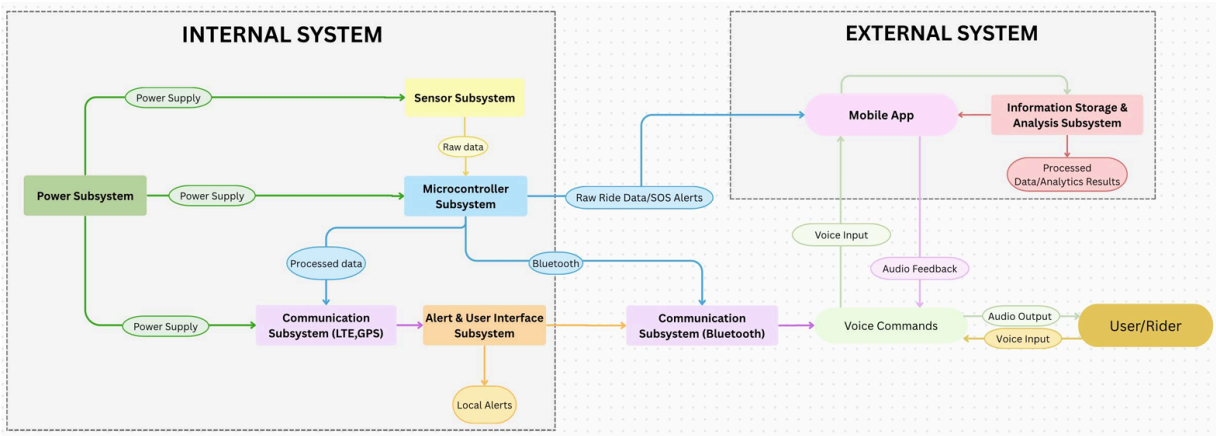

Figure 5: Block diagram of Rakib's internal and external system

Power Subsystem

This subsystem provides regulated power to all electronic devices that function under a lithium-polymer (LiPO) rechargeable battery. It is programmed with a built-in battery management system (BMS) that protects the battery from overcharging, short circuiting, etc., thereby optimizing its life [6]. The voltage regulator maintains supply to the sensors and microcontroller, and the system works for about 8 hours on a single charge, supplying power to the Sensor, Microcontroller, Communication, and Alert & User Interface subsystems.

Sensor Subsystem

The sensor subsystem gathers body signals, as well as motion data, from the rider. An accelerometer/gyroscope measures acceleration, angular velocity, and impact force, allowing the detection of unusual head motions or crashes. A MAX30102 PPG/HRV sensor measures the rider's heart rate and SpO_2, heart-rate variability, and oxygen levels in order to monitor for fatigue or stress. All the raw data is transmitted from the sensors to the microcontroller for further processing.

Microcontroller Subsystem

The microcontroller subsystem accepts the raw data input from all sensors for condition detection for fatigue and crash conditions. The microcontroller subsystem is the main control, data storage, processing, and logic component of Rakib, built around an ESP32 board. It analyzes the incoming raw data from the Sensor Subsystem and checks for fatigue and crash conditions. The processed ride data and SOS alerts are then sent to the Communication Subsystem (LTE/GPS), for forwarding to the app and cloud. Bluetooth from the Communication Subsystem (Bluetooth) allows for a local app interface and voice feedback. It then sends control signals to the Alert and User Interface Subsystem to actuate LED and buzzer alerts. Communication Subsystem The communication subsystem takes care of all wireless connectivity

required for normal operation and emergency response. There are two main communication pathways: LTE/GPS Communication Subsystem receiving processed data and SOS alerts from Microcontroller Subsystem sends them to external app and cloud database for location tracking and emergency response; and Bluetooth Communication Subsystem which gets data and commands from Microcontroller Subsystem and connects to the app for short-range communication. It gives audio feedback and status updates to User/Rider when the helmet is online, and takes voice commands from the app back to the helmet.

Alert and User Interface Subsystem

The Alert and User Interface Subsystem receive control signals from the Microcontroller and provides real-time feedback to the rider. It comprises LED indicators, a buzzer, a push button, and a speaker-microphone kit for communication through the Rakib mobile application. The LED displays system status: blue is for fatigue detected, red means SOS is not sent, while green shows the SOS is sent. Buzzer gives audible countdown for a short time before emission of SOS; it features a push button that can cancel SOS when it has been activated. The LED, buzzer, and push button should function offline to ensure that alerts and safety feedback are always available even without network connectivity.

Information storage and analysis subsystem

This subsystem is external and includes the mobile application/cloud database. It gets ride data and SOS alerts from the helmet through the LTE Communication Subsystem, storing them for long-term analysis. The app uses AI for the analysis found and updates the status by communicating the findings with the microcontroller via bluetooth to display the ride history, fatigue alerts, and safety logs. These features help in evaluating rider performance and improving fatigue detection accuracy over time.

3.2.2 System Characteristics

3.2.2.1 Functional/Performance Requirements

The accelerometer samples were taken at a frequency of 100 Hz, which is adequate for the recording of an active change in the acceleration or impact. Crash detection would significantly benefit from its ability to operate with minimal power consumption, despite the device being capable of providing sampling rates up to 1 kHz. MAX30102 PPG/HRV sensors sample data for a rate of 50Hz. MPU-6050 Accelerometer Sensor with linear acceleration as parameter has an accuracy of ±0.05 g and resolution of about 16 bits. MPU-6050 Gyroscope Sensor has as parameters Angular velocity, deriving its accuracy of ±0.2 °/s and resolution of about 16 bits. The impact detection and decision time will be under 300ms, while the SOS will be sent within 5s after the conformation of the crash. The Rakib helmet is expected to survive around 3 years of normal use, where each full charge lasts around 8 hours, and at least 80% of battery capacity is maintained after 300 full charging cycles, and Use of 4G LTE for real-time data upload and SOS

alert and uses GPS for live location tracking and route logging. Wi-Fi/Bluetooth connections are provided as backup for short-range data synchronization.

3.2.2.2 Physical Characteristics

Rider helmets must comply with all applicable ASTM F1446 safety standards to provide adequate impact absorption and resistance as per safety requirements. All integrated electronic components are securely encased in the helmet to avoid discomfort and are protected from damage to the helmet shell. Such design maintains the integrity of the helmet and safety while allowing for integrating sensors and modules without discomfort and offsetting the weight balance.

3.2.2.3 Electrical Characteristics

The battery is a LiPo rechargeable one that provides up to 8 hours of continuous operation for sensors and the AI model. Operating power is wisely restrained to 5W to maintain thermal safety and efficiency in power management. Bluetooth has been used for mobile connectivity and wireless data transmission between the helmet and the application.

The software subsystem is embedded and run through a web server that manages the reception of sensor data, AI feedback, and communication with the app. Its main purpose is to improve rider safety through intelligent data processing capable of detecting fatigue and crash conditions. Sensor data from MAX30102 PPG/HRV, accelerometer/gyroscope are collected, filtered, and processed locally and sent to the app to be used and logged. These local operations are intended to reduce latency and enable the detection to be accurate and fast acting in critical situations. TensorFlow Lite models conduct analysis on the data collected by the sensors and sent to the microcontroller to the web server. Then the feedback and results are sent back to the microcontroller for alerts and system response process. These AI models will not depend on any network connectivity, thus ensuring instant readiness during emergency conditions. Firebase used for storing logs and SOS records through the Rakib app. Cloud based storage enabled reliable data access and backups for long-term performance tracking. The app provides real-time warnings of fatigue, indicates crash events, and shows battery status. Users can, therefore, review their ride data and change system settings. A user interface to facilitate access to user-friendly information and continuous performance monitoring.

Chapter 4: Design, Standards, and Course Integration

4.1 Detailed Design

The Rakib AI Helmet is designed to protect motorcycle users via fatigue detection while also bettering emergency responses in the event of an impact. The system architecture includes a group of subsystems, specifically: Sensor, Microcontroller, Communication, Power, Alert, User Interface, Information Storage, and Analysis, totaling six subsystems. Physiological parameters, including heart rate variability, are monitored with the MAX30102 sensor. The MPU-6050 module is used to aid in the detection of shock, tilt, and irregular head movements at the center. Running the sensor data on-device with TensorFlow Lite AI models on the LilyGO T SIM7600G S3 (ESP32) microcontroller allows for millisecond-level response times. The system is also capable of operating in four modes: Normal Ride, Fatigue Detection, Crash or Emergency and Data Upload. Hence, it also generates alerts through its LEDs and buzzers or can create an SOS with the coordinates passed out through LTE/GNSS. Hardware testing has confirmed the functionality of the communication pipeline with respect to the push button, LED, and buzzer.

Application of engineering and scientific principles in all modalities including Electrical Circuit Theory ECEN 214 in power management design using Li Po battery and Battery Management System that applies Kirchhoff's Laws and source transformation for stable power and Electronics ECEN 325 where the performance design of signal conditioning for sensitive MAX30102 PPG and accelerometer data by filtering and amplifying the signals through operational amplifiers. ECEN 248 includes Digital Systems Design, backs control logics ensuring smooth transitions between states and alert mechanism with logic gates and sequential state machines. The AI model of fatigue detection uses mathematical and computer science principles to analyze the pattern of physiological and motion data for insights on predictive safety applications.

Compliant with the following standards; ASTM F1446 covers impact energy absorption, ISO 26262 deals with functional safety, and ISO/IEC 27001 covers data privacy. Several key challenges were identified; high power consumption, which was overcome by optimizing the Battery Management System to regulate power delivery based on the 3.7 V Li-Po battery and its maximum expected current draw ensuring up to 8 hours of uninterrupted operation. Also, it minimizes false alarms caused due to any kind of vibration or head movements, combining the input of various sensors. Environmental durability through an IP54 rating covers dust and water resistance, and a wide temperature range. Engineering strong innovations in security along with electrical, electronic, and digital systems make the Rakib helmet an excellent solution for safety. The integration of AI-driven fatigue detection and crash response mechanisms established by engineering standards can make them an advanced proactive accident prevention measure and rapid emergency assistance, this approach constitutes an advanced and relevant safety solution.

4.2 Technical Standards and Constraints

Health and Safety Standards

The Rakib AI Helmet has been built considering health and safety as the first concern, given that delivery riders can easily fall asleep or lose focus due to fatigue, leading to accidents. This helmet incorporates fatigue monitoring sensors-MAX30102 and an MPU 6050 accelerometer for crash detection to reduce such accidents and build post-ride data on analysis. For rigorous impact testing, it ensures compliance with the specification ASTM F1446, while for electronic system safety, it guarantees compliance with ISO 26262.

Hardware and Technical Standards

Critical constraints include data accuracy and power consumption, which are dealt with by the helmet's hardware and technical design. Multiple sensors assist in cross-verification, minimizing false warnings. A rechargeable Li-Po battery with an integrated battery management system provides approximately eight hours of operation. Power dissipation has been limited to a maximum of 3.7V, 1,400 mA to avoid overheating. Connectivity risks have been kept to a minimum by relying on LTE/GNSS coverage for SOS alerts. These measures ensure reliable performance under demanding conditions while maintaining efficiency.

Ethical and Social Standards

Ethical and social considerations are central to the design of the helmet. For privacy, the helmet secures all data with AES 256 encryption, compliant with ISO/IEC 27001 and thus secure for data transfer. Emphasis is placed on personalization, with each helmet made for one rider to ensure proper fit and accuracy, hence never shared. Cultural adaptation is also ensured, notably for Qatar food delivery riders for whom fatigue is a common concern. This ensures that the helmet is technologically advanced but socially responsive.

Environmental Standards

The helmet is made of hard-wearing materials that can sustain environmental conditions ranging from 10°C to 50° C. It also corresponds to the IP54 standard, protecting against dust and water entry. This will ensure that the helmet operates and is reliable in various climates and operation environments.

Economic Standards

Targeting a lower-cost prototype addresses economic considerations, and scalability and affordability are enhanced. The power optimization strategies include modular design for efficiency, making the helmet more competitive in comparison with existing commercial smart helmets. In this way, it balances affordability with innovation, making the solution more widely adoptable while not compromising on quality or safety.

4.3 Link Between ECEN Courses and the Project

4.3.1 ECEN 214: Electrical Circuit Theory

The Power Subsystem of the Rakib AI Helmet applies Electrical Circuit Theory directly where the rechargeable LiPo battery follows Battery Management System and is expected to provide stable power for continuous use. The analysis and energy management of the six subsystems apply principles such as Kirchhoff's Laws and source transformation. These techniques limit the maximum power draw (3.7V, 1.4kmA) thus improving efficiency, preventing thermal issues, and ensuring a stable voltage supply. The fuel gauge chip integration into the system for observing the battery's percentage contains circuit theory information that guarantees its dependable performance. These examples illustrate the application of ECEN 214 concepts which supports this reliability in this helmet.

4.3.2 ECEN 248: Introduction to Digital Systems Design

The control logic and real-time decision-making of the helmet are built upon fundamental digital system design principles. The ESP32 microcontroller implements sequential logic and state machines, concepts central to ECEN 248, to process sensor data and manage alert responses. Also, the structured logic flows were used to implement the four modes of the operations such as Normal, Fatigue and crash/emergency and Data Upload as described in chapter 1. The logic sequence of push button to led and the led to buzzer is detected by the Boolean algebra and the logic gates used guaranteeing the outputs will appear in a deterministic way upon the sensor inputs. This full framework of digital design allows the helmet to turn on the LED, buzzer and enable communication protocols in response to the rider conditions.

4.3.3 ECEN 325: Electronics

Some basic principles of electronics are used in this signal conditioning for the sensor data and the sensor like MAX30102 and the MPU will produce the raw signals that are needed for cleaning and stabilization in AI processing. The operational amplifier which is used to filter and amplify is lowering the noise and maintaining the precision of heart rate and acceleration. The linear circuit analysis and the AC circuit analysis will involve the phase of bode plots and are important in creating the concepts that are used in the design of first order and second order filters. These filters will make sure that the sensors output and their vibration including their interferences are being transmitted safely to the ESP32.

Integration Across Courses

The Rakib helmet exemplifies how foundational engineering disciplines converge into an integrated system. ECEN 214 (Circuit Theory) ensures efficient and safe power delivery, ECEN 248 (Digital Design) provides the robust control logic for system behavior, and ECEN 325 (Electronics) enables precise sensor interfacing and signal integrity. This multidisciplinary synthesis is critical for creating a real-time safety solution that is not only functionally robust but

also reliable under real-world operating conditions. The project demonstrates the practical application of coordinated engineering principles toward a common goal: enhancing rider protection through intelligent technology

Chapter 5: Project Itinerary

5.1 Current Progress

The Rakib project has successfully completed its foundational hardware phase (ECEN 403), validating the core electronic system. This phase ensured all major components, including the ESP32 controller, PPG and accelerometer sensor, are assembled, powered, and communicating effectively as a single unit. The crucial communication pipeline was validated by confirming real-time data transmission via Bluetooth, proven by successful live heart rate (HR) readings and instant accelerometer updates displayed on a nRF connect app. With the hardware foundation verified, the project transitions to the ECEN 404 software phase. This phase focuses on the full system validation plan, which includes developing the mobile application, implementing Cloud Backend AI/Analytics for fatigue monitoring, and conducting comprehensive system testing.

5.2 Progress Timeline and Validation Plan

The upcoming ECEN 404 phase will focus on software development, system integration, and performance validation. Key objectives include:

- Development of the Rakib mobile application for real-time monitoring and user interaction.
- Implementation of cloud-based AI analytics for fatigue detection and ride insights.
- Comprehensive system testing, including sensor accuracy, alert reliability, and power management under realistic riding conditions.

The validation plan will follow the structured timeline and task breakdown provided in Table 2, ensuring systematic progression toward a fully functional prototype.

Table 2: Validation Table

Status Indicators	
Completed	
On Schedule	
In Progress	

Behind Schedule	

Task	Deadline	Current Status
Team Formation & Mentor Confirmation	24 Aug 2025	
Literature Review & brainstorming	26 Aug 2025	
Proposal Design & Proposal	2 Sept 2025	
Identify Sensors & Main Components	6 Sept 2025	
Start Presentation #1 Preparation	10 Sept 2025	
Order Components	22 Sept 2025	
Presentation #1	29 Sept 2025	
Start drafting ConOps, FSR, Benchmarking, Milestone & Validation, Ethics reports	3 Oct 2025	
Components Received	22 Oct 2025	
Develop Theoretical & Mechanical Design	23 Oct 2025	
Final Draft ConOps, FSR, Benchmarking, Validation & Milestones, Ethics Reports	25 Oct 2025	
Component Testing / Programming	28 Oct 2025	

Circuit Design	1 Nov 2025	
Final Presentation Preparation	2 Nov 2025	
Interface Microcontroller Analog Sensors	3 Nov 2025	
Interface Microcontroller Digital Sensors	5 Nov 2025	

Communication & Alert Signal Testing	8 Nov 2025	
Microcontroller Response Validation	10 Nov 2025	
Develop Power System	12 Nov 2025	
Helmet Enclosure Assembly	14 Nov 2025	
Gather Databases & Relevant Resources for AI Model	19 Nov 2025	
Develop Fatigue & Relevant Resources for AI Model	21 Nov 2025	
System Enclosure Finalization	22 Nov 2025	
Final Presentation	26 Nov 2025	
Hardware Testing	3 Dec 2025	

5.3 Conclusion

The Rakib AI Helmet represents a major advancement in motorcycle safety through the integration of real-time fatigue monitoring and automated crash detection. This successful prototype development has combined AI-driven physiological sensing with proactive alerting and emergency response for the very first time. Rakib addresses a very critical gap in rider safety, particularly that of delivery personnel in high-risk environments, by combining sensor fusion, embedded AI, and robust communication subsystems within a standard helmet shell.

Future work will concern software integration, refining the AI models for better accuracy, and power optimization in order to extend battery life. Further, real-world data is to be collected and the system validated in preparation for safety certification and possible deployment. When finally realized, Rakib has the potential to complement rider safety with accessible, efficient, and intelligent protection, contributing to lower accident rates and better well-being among motorcyclists.

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